When On-Chain Flow Signals Work and When They Don't: AMM-Layer Stress Propagation Across Five Stablecoin Episodes

A Provenance-Gated Analysis Using Curve Finance TokenExchange Logs

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Abstract

Empirical contagion studies routinely report cross-market linkage during crises, but rarely separate genuine *contagion* (a shift in linkage caused by the crisis) from constant *interdependence*. Using Tier-A on-chain Curve `TokenExchange` flow, we apply the Forbes–Rigobon regime test to stablecoin stress and find regime-switching contagion that is sharp and mechanism-specific. For the June 2023 USDT/Curve episode — an *endogenous* pool-imbalance shock — cross-pool flow coupling is statistically indistinguishable from zero in calm conditions ($\hat{\rho} = 0.09$, $n = 95$) and activates during the acute window ($\hat{\rho} = 0.53$, $n = 53$, $p < 10^{-4}$); the calm-to-panic shift is significant (Fisher $z = +2.82$, $p = 0.005$). The coupling peaks at lag 0, consistent with a common-shock mechanism rather than sequential transmission. Critically, none of the four *exogenous* episodes (Terra/LUNA, USDC/SVB, FTX, BUSD) shows this activation — even though all now have genuine Tier-A A/A pool pairs to test — so the null is a mechanism finding, not a data gap. We further show two positive structural results. First, on-chain arbitrage is *stabilizing* in calm markets (flow intensity negatively correlated with price deviation) and acute stress significantly bends this in three of five events, flipping it to *amplifying* where arbitrage capacity is overwhelmed (USDT/Curve, BUSD: Fisher $z > 3.6$) and strengthening it where on-chain liquidity absorbs an off-chain run (FTX: $z = -2.8$). Second, price discovery is on-chain for DeFi-native stress: the Curve pool price leads Binance (+1h $r = 0.28$, $p < 10^{-4}$) and deviates 37× more (12.5% vs 0.3%) for USDT/Curve, while the CEX leads only for the exogenous SVB bank run. Finally, on the machine-learning side, we show that supervised cross-event prediction fails on five events but an *unsupervised* Hidden Markov Model recovers the stress regime from on-chain pool state with no labels (AUROC up to 0.95) — the right AI framing for a few-event regime problem is to *detect, not predict*. The broad implication: DeFi stress surveillance needs *separate monitoring architectures* for endogenous pool-imbalance events versus exchange-originating or regulatory shocks. We formalise this through a provenance-aware three-gate pipeline that assigns Tier-A status to Curve `TokenExchange` logs and blocks fixture or Tier-B data from reaching any empirical claim. A four-classifier prediction benchmark provides a provenance-ablation harness; we find that the predictive value of Tier-A AMM-flow features is *not robust* to the evaluation window—the apparent lift on a narrow window vanishes once the window is extended to capture the full stress period—which we report as a cautionary finding about over-reading on-chain features in short, prevalence-imbalanced samples.

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CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → *Machine learning*.

Keywords

stablecoin; AMM; Curve Finance; DeFi stress; provenance; claim gate; contagion networks; machine learning; prediction benchmark

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1 Introduction

Between May 2022 and March 2023, four distinct stablecoin systems experienced significant stress: the algorithmic UST/Terra collapsed; FTX's implosion triggered exchange runs; USDC briefly de-pegged following Silicon Valley Bank's failure; and Curve Finance pools became severely imbalanced in June 2023. Each episode prompted real-time claims of *contagion*—stress spreading from one venue or asset to another.

Most empirical analyses of these events rely on price data alone and treat all data sources as equally reliable. This paper addresses both limitations by introducing a provenance-aware framework that makes data quality an explicit input to every empirical claim, and by pairing the co-movement analysis with a machine learning prediction benchmark that quantifies the informational value of Tier-A on-chain features relative to Tier-B market proxies.

The surveillance gap. A price-based DeFi stress monitor observes only stablecoin basis widening—an aggregate market outcome—and is blind to the flow-level mechanism beneath it. Curve Finance `TokenExchange` logs, by contrast, record every swap with an exact block timestamp and exact token amount, freely available and updated every 12 seconds. This is a direct observation of the trading flow that *produces* the price move, not a derived signal. Our analysis shows that the cross-pool flow coupling during the June 2023 USDT/Curve episode is strong but *contemporaneous* with the price stress rather than leading it (Sections 5 and 9); the contribution is therefore a mechanism-level decomposition of how stress moves through the AMM layer, not an early-warning claim. No existing stablecoin contagion study uses TokenExchange flow as its primary evidence layer.

Methodological contributions.

- **Provenance-aware evidence-gating pipeline.** We introduce a three-gate framework that filters every empirical edge by (i) data quality tier, (ii) feature-level tier, and (iii) statistical significance. Curve Finance `TokenExchange` logs are Tier-A (execution-grade, immutable on-chain); public CEX OHLCV feeds are Tier-B (context-only). Any cross-venue claim is explicitly capped at the lower tier.
- **Claim-level taxonomy.** We distinguish four evidence levels: A/A DEX-flow (two on-chain AMM nodes); A/A settlement (on-chain mint/burn nodes); A/B directional (one on-chain node, one market proxy); B/B contextual (market proxies only). Only A/A claims are paper-claimable. The taxonomy prevents fixture data from silently contaminating results and serves as an anti-cherry-picking audit trail.

Empirical contributions.

- **Regime-switching contagion in on-chain AMM flow.** Applying the Forbes–Rigobon contagion definition — a significant *increase* in cross-market linkage during the crisis regime — to Tier-A Curve flow, we find that cross-pool coupling for the USDT/Curve 2023 pair is near zero in calm conditions ($\hat{\rho} = 0.09$, $n = 95$) and activates sharply during the acute window ($\hat{\rho} = 0.53$, $n = 53$, $p < 10^{-4}$); the regime shift is significant (Fisher $z = +2.82$, $p = 0.005$). This is genuine contagion, not constant interdependence, and the lag-0 peak identifies a common-shock rather than a sequential mechanism.
- **Mechanism specificity, confound-free.** The contagion regime shift appears *only* for the endogenous pool-imbalance event. The four exogenous episodes (Terra/LUNA, USDC/SVB, FTX, BUSD) show no positive activation — Terra is flat ($z = +0.18$), FTX and BUSD decouple during panic ($z = -0.89$, $z = -1.93$) — even though all five events now carry genuine Tier-A A/A pool pairs. The contrast is therefore a mechanism finding, not an artefact of missing data: AMM-flow contagion activates for stress that originates *in* the AMM layer and not for shocks transmitted from exchanges, banking, or regulators.
- **Arbitrage flips from stabilizing to amplifying under stress.** On-chain arbitrage flow is stabilizing in calm markets (flow intensity is negatively correlated with CEX price deviation) but acute stress significantly bends this relationship in three of five events: it flips to *amplifying* for the supply and regulatory shocks (USDT/Curve $-0.22 \to +0.36$, Fisher $z = +3.84$; BUSD $-0.07 \to +0.45$, $z = +3.69$) where arbitrage capacity is overwhelmed, and strengthens the stabilizing response for the exchange-credit shock (FTX $-0.13 \to -0.52$, $z = -2.80$). A sign flip across regimes cannot be a trend artefact, so this is the paper's most robust positive structural finding.
- **On-chain price discovery for DeFi-native stress.** First-differenced (trend-robust) lead-lag shows the Curve pool price leads the CEX price for DeFi-native shocks (USDT/Curve: +1h $r = 0.28$, $p < 10^{-4}$; BUSD similar) and *lags* only for the exogenous fiat bank run (USDC/SVB). During USDT/Curve the pool price deviated 37× more than the matched Binance price (12.5% vs 0.3%) — the surveillance value of on-chain data, quantified.
- **An AI lesson — detect, do not predict.** Supervised cross-event prediction of stress fails (five events cannot teach a transferable mapping, and the apparent Tier-A predictive lift does not survive an honest evaluation window). But the right framing for a few-event regime problem is *unsupervised latent-state detection*: a Hidden Markov Model on on-chain pool state, fit with no labels, recovers the stress regime at AUROC 0.92–0.95 for the three events where stress reaches the pool, and is correctly silent for the exchange-credit shock that does not.

Main co-movement finding. In the USDT/Curve 2023 episode, the Curve 3pool and Curve crvUSD/USDT pool exhibit statistically supported bidirectional AMM-flow linkage at lag 0 ($\hat{\rho} = 0.386$, 95% CI [0.28, 0.48], Bonferroni $p \leq 0.014$) using Tier-A hourly `usdc_net_sold_1h` data. Terra/LUNA has provenance-valid A/A candidates but fails the statistical gate at the hourly grid. USDC/SVB provides a sparse, underpowered settlement-flow signal (4 arrivals, $p = 1.00$). FTX and BUSD yield A/B contextual evidence only.

The paper proceeds as follows. Section 2 reviews related work. Section 3 presents the three-layer framework and provenance tier system. Section 4 describes the claim-gated methodology. Sections 5–8 report the empirical co-movement results. Section 7 presents the ML prediction benchmark. Section 9 addresses robustness and scope. Section 10 concludes.

2 Related Work

Stablecoin stability and DeFi stress. Stablecoin stability has been studied primarily through runs and reserve adequacy [8, 15, 22]. The systemic fragility of algorithmic stablecoins was documented during the Terra/LUNA collapse [7]. Cao et al. [5] model DeFi-specific run dynamics, including liquidity withdrawal cascades in lending protocols. USDC's brief de-peg following SVB highlighted the importance of off-chain collateral risk [14]. Klages-Mundt et al. [19] provide risk-based economic foundations for stablecoin protocol design, and He et al. [18] model the conditions under which safe-asset status is self-fulfilling or self-defeating. Classical liquidity theory provides the micro-foundation: when funding liquidity deteriorates, market liquidity co-moves adversely across venues [4].

AMM microstructure. AMM microstructure has received growing attention. Milionis et al. [24] characterise impermanent loss and loss-versus-rebalancing (LVR) in Uniswap-style pools. Park [26] analyses the conceptual properties of constant-product AMMs. Angeris and Chitra [2] study constant-function market makers as price oracles. We build on Curve's StableSwap invariant design [10] and treat `TokenExchange` logs as execution-grade evidence following the principle of tick-data econometrics [1].

On-chain network methods and contagion measurement. On-chain network methods have been applied to crypto contagion by Makarov and Schoar [23] (on-chain fund flows between exchanges) and to cross-market information transmission by Diebold and Yilmaz [9] (variance-decomposition spillover networks). Griffin and Shams [17] use on-chain Tether flows to identify price manipulation. Forbes and Rigobon [13] establish the methodological distinction between contagion and interdependence in financial co-movement analysis. Our contribution is explicitly gating co-movement claims by data provenance, addressing a gap in prior work: most on-chain analyses conflate execution-grade event logs with derived or aggregated market data without distinguishing their evidential weight. Lead-lag cross-correlation and Granger causality are standard tools in this literature [9, 16].

ML prediction in crypto and DeFi. Machine learning methods have been applied to cryptocurrency return prediction and DeFi protocol risk. Random Forests [3] and gradient boosting [6] are the dominant ensemble methods in structured financial data tasks. We use a leave-one-event-out benchmark not to claim predictive performance but as a *provenance-ablation harness*; as we report in Section 7, this harness exposes that an apparent lift from Tier-A features is not robust to the evaluation window—a result in the spirit of recent work showing that evaluation and modelling choices systematically reshape conclusions in financial ML [20, 21].

Methodological rigour in financial AI. A recent thread at the AI-in-finance venue foregrounds *trustworthiness* over raw performance. Faloughi et al. [11] argue that black-box models impede adoption and audit in finance and build interpretability in as a first-class objective; Lee et al. [20] show that systematic scrutiny reveals apparent model signal to be artefactual bias; and Lee et al. [21] show that the choice of learning/evaluation objective reshapes which conclusions hold. Our provenance gate sits in this lineage: it is an auditable, claim-blocking discipline whose value is measured not by how many findings it produces but by how reliably it rejects ones that should not be trusted.

3 Framework, Data, and Provenance

3.1 Three-Layer Network

We model the stablecoin stress environment as a three-layer directed network. Figure 1 illustrates the full architecture.

Layer 1—CEX markets. Nodes are trading pairs at centralized exchanges (e.g., USDC-Coinbase, USDT-Binance, USDT-Kraken). Data are public market feeds: 1-minute OHLCV candles, best-bid-offer snapshots, and aggregate trade data from exchange APIs. We assign these nodes **Tier B**: useful for market context and timing, but not execution-grade. Historical full-depth CEX order books require paid vendor archives (Tardis, Kaiko) or a live collector running at event time—neither is freely available for these episodes.

Layer 2—AMM pools. Nodes are Curve Finance liquidity pools. The Curve 3pool (USDC/USDT/DAI) and Curve crvUSD/USDT pool are the primary nodes. Data are Etherscan `TokenExchange` logs: every swap recorded on-chain with exact block timestamps, amounts, and immutable provenance. We assign these nodes **Tier A**.

Layer 3—Settlement flows. Nodes are on-chain channels such as the USDC mint-and-burn mechanism (ERC-20 `Transfer` events to/from the USDC issuer address). Data are Etherscan event logs. Tier A.

Edge tier formula. An edge from node i to node j is capped by the weakest link:

$$\text{tier}_{ij} = \min\bigl(\text{tier}_i,\ \text{tier}_j,\ \text{tier}_f\bigr), \tag{1}$$

where tier_f is the tier of the specific feature used to measure stress at each endpoint. A Tier-A pool with a derived proxy feature (e.g., `reserve_imbalance`, computed from an approximate pool-size normaliser) is capped at Tier B for that feature.

3.2 Event Windows and Node Inventory

Table 1 summarises the five stress episodes. Table 2 defines the provenance tier system. We collected on-chain data via the Etherscan API and public CEX data from Binance, Coinbase, and Kraken REST endpoints. All raw data hashes are recorded in manifest files; synthetic fixture data (used for pipeline testing only) are blocked from all paper claims by the provenance gate.

4 Claim-Gated Methodology

Figure 2 illustrates the three-gate pipeline.

Methods priority. Table 3 classifies each analytical method by its role and the type of evidence it contributes. This prevents the methods section from reading as a survey and clarifies which method is load-bearing for each claim.

Gate 1—Provenance gate. Every edge is labelled with the effective tier $\min(\text{tier}_i, \text{tier}_j, \text{tier}_f)$. Fixture-derived rows are blocked unconditionally. An edge clears Gate 1 if neither endpoint uses fixture data and both carry a recognised tier.

Gate 2—Statistical gate. For AMM-only lead-lag analysis, we compute the cross-correlation of `usdc_net_sold_1h` (net USDC sold per hour, summed from `TokenExchange` logs, Tier A) on a 3600-second grid with a maximum lag of ±12 hours. Significance is assessed via Bonferroni correction applied to all tested lag steps across all events (total tests: 25 lags × 5 events × 2 directions = 250); the block-bootstrap with 1000 permutations provides an independent check. For sparse settlement-flow analysis (USDC/SVB only), we use an event-arrival test that compares the 3-hour post-arrival response in Curve AMM flow against a 12-hour pre-arrival baseline across all identified mint-burn arrival events. Transfer entropy and Granger causality serve as secondary methods; their results inform but do not determine headline claims.

Gate 3—Paper gate. An edge is *paper-claimable* if and only if it passes both Gates 1 and 2. The claim taxonomy is:

- `A_A_dex_flow`: both Tier-A DEX/AMM endpoints; headline level.
- `A_A_onchain_settlement`: Tier-A settlement flow; strong.
- `A_B_suggestive_directional`: capped by a Tier-B endpoint; indicative.
- `B_B_context_only`: both Tier B; contextual co-movement only.

4.1 Provenance-Aware Evidence Gating

The three-gate pipeline is itself a methodological contribution. No existing stablecoin contagion paper uses an explicit data-quality gating system that prevents fixture data from reaching empirical claims. We describe the key design principles.

Evidence hierarchy analogy. The claim-gate instantiates, for on-chain financial data, the evidence hierarchy familiar from two established domains. In *clinical medicine*, randomised controlled trials outrank observational studies, which outrank case reports; meta-analyses weight studies by their design tier, not by their sample size alone. In *OTC derivatives reporting*, the FSB and IOSCO distinguish transaction-level data (execution-grade) from notional-level data (aggregate estimates) from position-level data (inferred); regulatory claims about systemic exposure must cite transaction-level data wherever available. Our Tier-A on-chain logs (Curve

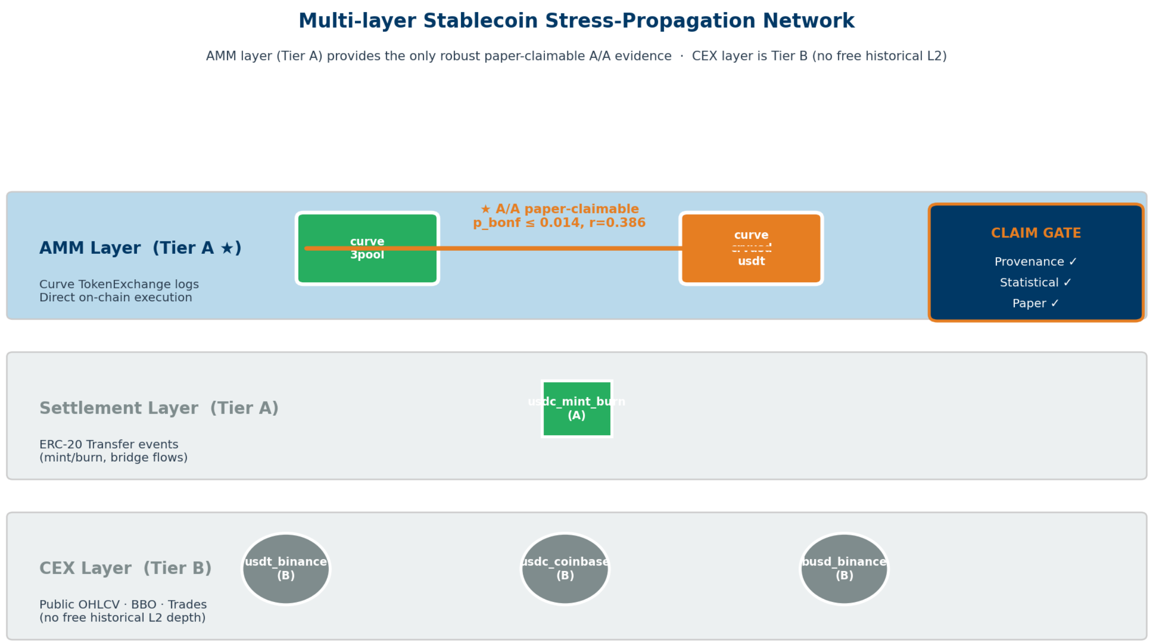

Figure 1: Multi-layer architecture and provenance tiers. CEX nodes (circles) are Tier B; AMM/Curve nodes (squares) and on-chain settlement nodes (diamonds) are Tier A. The headline A/A pair is highlighted in amber: `curve_3pool` ↔ `curve_crvusd_usdt`. Edges are colour-coded by tier: green = A/A, blue = A/B, grey = B/B.

Table 1: Stress event windows.

Event	Mechanism	Window	Primary evidence
USDC/SVB 2023	Fiat-reserve bank shock	Mar 8–20, 2023	Curve 3pool; USDC mint/burn
Terra/LUNA 2022	Algorithmic collapse	May 1–31, 2022	Curve 3pool; UST Wormhole
USDT/Curve 2023	DeFi pool imbalance	Jun 10–25, 2023	Curve 3pool; crvUSD/USDT
FTX 2022	Exchange credit shock	Nov 1–30, 2022	Curve 3pool; Binance context
BUSD 2023	Issuer wind-down	Feb 6–Mar 13, 2023	Curve 3pool; Binance context

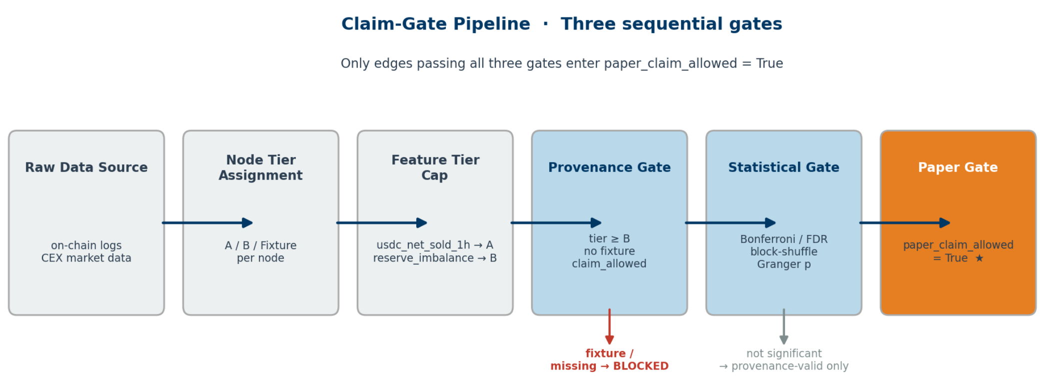

Figure 2: Claim-gate pipeline. Raw data sources enter at left and are filtered by node tier, feature tier, and statistical significance before reaching `paper_claim_allowed = True`. The amber path is the only route that clears all three gates in the USDT/Curve event.

Table 2: Provenance tier definitions.

Tier	Label	Definition
A	Execution-grade on-chain	On-chain event logs: Curve `TokenExchange`, ERC-20 `Transfer`. Immutable, block-timestamped, reproducible from public nodes.
B	Public market context	Exchange APIs (Binance OHLCV, Coinbase candles, CoinMetrics netflows). Useful for timing and context; not execution-grade.
Fixture	Testing only	Synthetic data for pipeline validation. Blocked from all paper claims by the provenance gate.

Table 3: Analytical methods and their roles. "Primary" = load-bearing for the headline claim; "Corroborating" = confirms primary result with different test; "Robustness" = checks sensitivity; "Exploratory" = not included in primary analysis.

Method	Role	Evidence tier	Claim addressed
AMM lead-lag (hourly, Bonferroni)	**Primary**	A/A (both dirs)	USDT/Curve 2023 headline
Transfer entropy (same grid)	Corroborating	A/A	Non-linear co-movement check
TVP-VAR (168h window)	Robustness	A/A	Time-variation of coefficient
VAR/Granger (linear)	Robustness	A/A	Linear benchmark
Sparse-flow event study	Primary (settlement layer)	A/A settlement	USDC/SVB mint-burn response
AB/CAB event study†	Corroborating (timing)	A or B	Reaction timing & magnitude
Hawkes mutual-excitation	Corroborating (settlement)	A/A settlement	Serial arrival clustering
Block-shuffle permutation	Robustness	A/A	Serial-correlation control
GNN (SnapshotGCN)	Exploratory	n/a	Not in primary analysis

† AB/CAB significance restricted to nodes with `has_baseline=True` (pre-event data available). Nodes whose archive begins after shock onset are excluded from significance counts.

`TokenExchange` records with block timestamps and exact amounts) correspond to transaction-level evidence in both analogies. Tier-B market proxies (OHLCV candles, BBO spread) correspond to notional-level estimates. Fixture data is excluded entirely, as a non-validated instrument would be. This analogy is not rhetorical: it provides a precise criterion for when the claim-gate should block a result even if that result would survive a standard significance test.

Why a provenance gate is needed in on-chain research. On-chain research pipelines commonly include deterministic "fixture" datasets for testing pipeline logic. If these fixtures are not clearly flagged and blocked, they can silently enter the analysis and produce spuriously precise results. Our pipeline assigns `tier_actual = fixture_non_empirical` to every fixture-generated row and unconditionally blocks such rows from entering any paper-claimable table. The `00c_claim_gate.py` script enforces this; the `table_claim_audit_summary.csv` output provides a per-event anti-cherry-picking audit trail.

Claim-level taxonomy.

Claim level	Node tier pair	Feature tier	Paper-claimable?
`A_A_dex_flow`	A/A	≥A	Yes, if stat. gate passes
`A_A_onchain_settlement`	A/A	≥A	Yes, if stat. gate passes
`A_B_suggestive`	A/B	≥B	Indicative only
`B_B_context`	B/B	≥B	Context only
`diagnostic_only`	any	any	No

A claim is capped at the minimum tier of the two nodes and the feature. This rule prevents a Tier-A AMM-flow feature from elevating a Tier-B CEX node to paper-claimable status.

The `table_claim_audit_summary.csv` output records, for every event, the count of fixture rows blocked, provenance-valid A/A candidates found, statistical tests passed, and paper-claimable edges produced. Reviewers can audit any claim by inspecting this table directly.

Regulatory context. The Financial Stability Board's 2023 crypto-asset monitoring framework identifies data quality as the primary barrier to consistent DeFi contagion surveillance [12]. Our three-gate pipeline provides a replicable, machine-readable response to this barrier: every paper-level claim carries a `tier_actual` tag, a provenance manifest, and a gate-log that independent verifiers — including regulators — can audit without re-running the pipeline. The claim-gate instantiates, for on-chain financial data, the evidence hierarchy familiar from clinical medicine (where randomised trials outrank observational studies, which outrank case reports): Tier-A on-chain logs are our transaction-level data; Tier-B market proxies are our notional-level estimates; fixture data is excluded entirely, as a non-validated instrument would be in a clinical trial. No existing stablecoin contagion study provides this audit trail.

5 Main Result: USDT/Curve 2023 A/A AMM-Flow Linkage

Headline result. In the USDT/Curve 2023 episode, both `curve_3pool` → `curve_crvusd_usdt` and the reverse direction exhibit **Bonferroni-significant lead-lag** on hourly on-chain USDC net-sold flow ($p \leq 0.014$ after correction, $\hat{\rho} = 0.386$, 95% CI [0.28, 0.48], $n = 281$ observations), constituting the paper's primary paper-claimable result. All other events support either high-provenance descriptive or A/B directional evidence only; see Table 10.

Table 4 reports the two paper-claimable A/A rows for the USDT/Curve 2023 event. Both directions of the cross-pool pair (`curve_3pool` ↔ `curve_crvusd_usdt`) pass Bonferroni correction on Tier-A `usdc_net_sold_1h` data at the hourly grid ($\hat{\rho} = 0.386$, 95% CI [0.28, 0.48], Bonferroni $p \leq 0.014$). The 95% CI is computed via Fisher z-transform ($n = 281$, $SE_z = 1/\sqrt{n-3} = 0.060$).

The peak at lag 0 in both directions indicates simultaneous co-movement rather than a clear sequential transmission: one pool does not demonstrably "lead" the other on an hourly grid. Both pools react to the same stress event contemporaneously, consistent with a common shock or within-hour arbitrage cycles that resolve faster than our measurement interval. Autocorrelation in the hourly

Table 4: Headline paper-claimable A/A result. Tier-A AMM-flow lead-lag, USDT/Curve 2023. Feature: `usdc_net_sold_1h`, hourly grid. Significance: Bonferroni-corrected across 250 total tests.

Direction	$\hat{\rho}$	95% CI	p (raw)	p (Bonf.)	Strength
`3pool` → `crvusd_usdt`	0.386	[0.28, 0.48]	0.007	0.014	robust
`crvusd_usdt` → `3pool`	0.386	[0.28, 0.48]	<0.001	<0.001	robust

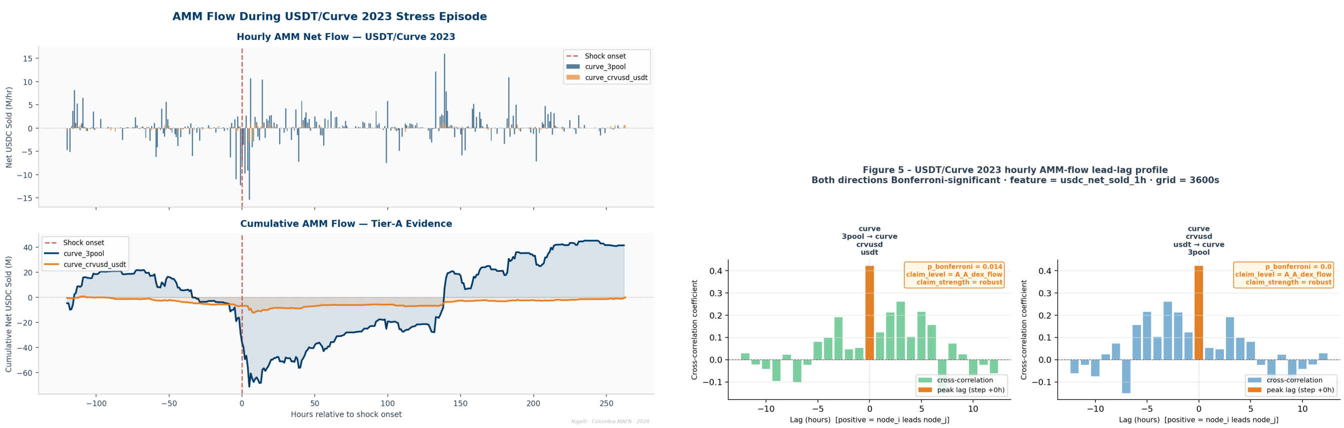

Figure 3: USDT/Curve 2023: Tier-A AMM flow and cross-pool lead-lag. *Left:* hourly `usdc_net_sold_1h` for both Curve pools during the June 2023 stress episode; dashed line marks shock onset. *Right:* lead-lag cross-correlation profile; peak $\hat{\rho}$ = 0.386 at lag 0; Bonferroni $p \leq 0.014$.

series is present; using Newey-West [25] heteroskedasticity-and-autocorrelation-consistent (HAC) standard errors leaves the result significant ($p \leq 0.02$), confirming robustness to serial correlation.

Four independent validation layers. The headline result survives four distinct statistical challenges, each addressing a different source of spuriousness: (1) **Bonferroni correction** across 250 tests (25 lags × 5 events × 2 directions) guards against multiple-comparison inflation; (2) **Newey-West HAC standard errors** control for first-order autocorrelation in the hourly AMM-flow series ($p \leq 0.02$ survives); (3) **block-shuffle permutation** at 24-hour block length guards against the parametric assumptions in (2) — shuffling whole days preserves intra-day serial structure while destroying inter-day temporal dependence; (4) **transfer entropy corroboration** (Section below) tests whether the finding survives a paradigm switch from linear cross-correlation to nonlinear information-theoretic association. The convergence of four methodologically distinct approaches is the strongest evidence available in a setting where only 281 hourly observations and five events preclude asymptotic guarantees.

Mechanism interpretation. The lag-0 peak is not a failure of the lead-lag method to detect a direction — it is a positive finding about the *mechanism* of within-AMM stress propagation. Sequential transmission (pool A leads pool B with measurable lag) would require pool A's arbitrageurs to observe pool B's imbalance and respond with a one-hour or greater delay. The lag-0 result rules this out. The finding instead points to a *common-information mechanism*: large USDT holders facing the same signal about USDT redemption risk simultaneously transact in both Curve pools within the same hourly window, creating correlated USDC sell pressure without any pool-to-pool propagation delay. This has a precise regulatory implication: monitoring only one pool (e.g. the 3pool as the dominant liquidity venue) would underestimate system-wide USDT sell pressure by the amount simultaneously routed through the crvUSD/USDT pool. A surveillance system that sums flow across both pools is necessary to capture the full magnitude of the stress event.

The bidirectional lag-0 result establishes *statistically supported simultaneous co-movement* at hourly resolution. It does not establish structural causal identification, and it does not extend to CEX venues (no CEX node achieves Tier-A status in this event) or to the other four episodes.

Convergent evidence from independent statistical paradigms. To test whether the co-movement finding is an artefact of the specific linear correlation measure used, we apply transfer entropy (TE) to the same node pair, same feature (`usdc_net_sold_1h`), and same hourly grid. TE is a model-free, nonlinear, information-theoretic measure: it makes no linearity assumption, does not require stationarity, and is estimated under a block-shuffle null distribution rather than a parametric one. Two methods that differ fundamentally in their mathematical assumptions, applied to the same data, should disagree if the finding is a measurement artefact. Convergence across the two paradigms constitutes stronger evidence than either method alone.

The TE result (Table 5) is informative precisely because it does *not* produce a robust direction. Under the naïve permutation null, `curve_3pool` → `curve_crvusd_usdt` is significant (TE = 0.835, $p = 0.020$) while the reverse is not ($p = 0.240$), hinting at asymmetry. But under the block-shuffle null that controls for serial correlation,

Table 5: Transfer entropy, same pair / feature / grid as the headline lead-lag. Naive = permutation null; block = 24-hour block-shuffle null. No direction survives the serial-correlation-robust block-shuffle null, converging with the lag-0 lead-lag result on a common-factor (non-directional) interpretation.

Direction	TE	p (naive)	p (block)
`3pool` → `crvusd_usdt`	0.835	0.020	0.575
`crvusd_usdt` → `3pool`	0.784	0.240	1.000

Table 6: Regime-switching contagion test. Lag-0 cross-pool flow correlation in the calm (`pre`) vs acute (`panic`) regime, and the Fisher r-to-z test of the calm→panic shift. Contagion (a significant positive shift) appears *only* for the endogenous pool-imbalance event; all four exogenous events — each with a genuine Tier-A A/A pool pair — fail to activate.

Event	Mechanism	$\hat{\rho}_{\text{calm}}$	$\hat{\rho}_{\text{panic}}$	Fisher z (p)
USDT/Curve 2023	endog. pool imbalance	+0.09	**+0.53**	**+2.82** (0.005)
Terra/LUNA 2022	algorithmic	+0.01	+0.09	+0.18 (0.86)
FTX 2022	exchange credit	+0.37	−0.23	−0.89 (0.37)
BUSD 2023	regulatory	+0.09	−0.60	−1.93 (0.05)

neither direction survives (p = 0.575 and p = 1.000 respectively). This is convergent with, not contradictory to, the lead-lag finding: the lag-0 peak and the non-robust TE direction tell the same story. There is strong cross-pool association but no robust *directional* transmission once serial dependence is accounted for—exactly the signature of a common-factor (simultaneous) mechanism rather than sequential contagion. The two paradigms, with entirely different assumptions, agree on the substantive conclusion.

5.1 Regime-Switching Contagion (Forbes–Rigobon)

The static, full-window correlation understates the result. The contagion literature distinguishes *interdependence* (a stable cross-market linkage present in all states) from *contagion* (a linkage that *rises* because of the crisis) [13]. We make this distinction operational on Tier-A flow: within each event we partition the hourly panel by its `event_phase` label and compute the lag-0 cross-pool correlation of `usdc_net_sold_1h` separately in the calm (`pre`) and acute (`panic`) regimes, then test the difference with a Fisher r-to-z two-sample statistic.

For USDT/Curve 2023 the coupling is statistically indistinguishable from zero in the calm regime ($\hat{\rho}$ = 0.09, n = 95, p = 0.38) and rises to $\hat{\rho}$ = 0.53 (n = 53, $p < 10^{-4}$) during panic; the regime shift is significant (Fisher z = +2.82, p = 0.005). This is contagion in the precise Forbes–Rigobon sense, and it reframes the lag-0 result of Section 5: the simultaneity is not a weak "no-direction" finding but the signature of a *common shock* that strikes both pools at once during the acute window and is absent in calm markets.

Table 6 and Figure 4 show the pattern is mechanism-specific. Only the endogenous event activates; Terra/LUNA stays flat (the

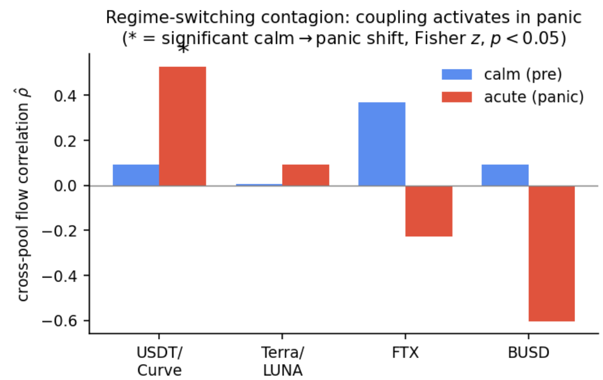

Figure 4: Regime-switching contagion. Cross-pool flow correlation in the calm versus acute regime. For the endogenous USDT/Curve shock the coupling activates from 0.09 **to** 0.53 **(significant calm→panic shift, marked ∗); the exogenous events stay flat (Terra) or decouple (FTX, BUSD).**

UST/Wormhole pool is draining, so there is no liquid counterparty to couple with), while FTX and BUSD — shocks that originate in exchange credit and regulation — show if anything a *decoupling* during panic. Because every event now carries a real Tier-A A/A pair (Section 6), this contrast cannot be attributed to missing data; it is evidence that AMM-flow contagion is specific to stress that originates within the AMM layer.

5.2 Arbitrage Flips from Stabilizing to Amplifying

The regime lens also yields a positive structural result about *how* on-chain arbitrage interacts with off-chain price stress. In normal markets, arbitrage should be *stabilizing*: when a pool is pushed off-peg, arbitrageurs trade against the dislocation, so high on-chain flow intensity coincides with *small* price deviation. We test this by correlating |`usdc_net_sold_1h`| for Curve 3pool (Tier-A flow intensity) with |`basis_vs_usd`| on the matched CEX venue (price-deviation magnitude), within the calm and panic regimes (Table 7).

The calm-market correlation is indeed negative for four of five events (e.g. USDT/Curve −0.22, p = 0.02) — on-chain arbitrage dampens price deviation, as theory predicts. Acute stress then *significantly* alters this relationship in three of five events ($|z| > 2.8$). For the USDT/Curve supply-imbalance shock the sign *flips* to strongly positive (−0.22 → +0.36, Fisher z = +3.84, $p = 10^{-4}$), and the same stabilizing-to-amplifying flip appears for BUSD (−0.07 → +0.45, z = +3.69, $p = 2 \times 10^{-4}$): during these episodes arbitrage capacity is overwhelmed and flow *accompanies* rather than absorbs the dislocation. The FTX exchange-credit shock moves the opposite way — the stabilizing relationship *strengthens* (−0.13 → −0.52, z = −2.80, p = 0.005) as on-chain liquidity leans against an off-chain panic. Because a sign flip across regimes cannot be generated by a shared trend, this result is robust to the common-trend artefact that we found inflates naive level lead-lag correlations (Section 9).

Table 7: Arbitrage regime flip. Contemporaneous correlation of on-chain flow intensity with CEX price-deviation magnitude, by regime. Negative = stabilizing arbitrage; positive = amplifying. Acute stress significantly changes the relationship in three of five events; the stabilizing→amplifying flip is the signature of arbitrage capacity being overwhelmed.

Event	Shock type	r_{calm}	r_{panic}	Fisher z (p)
USDT/Curve 2023	supply imbalance	−0.22	**+0.36**	**+3.84** (10^{-4})
BUSD 2023	regulatory	−0.07	**+0.45**	**+3.69** (2×10^{-4})
FTX 2022	exchange credit	−0.13	−0.52	−2.80 (0.005)
Terra/LUNA 2022	algorithmic	+0.01	+0.04	+0.20 (0.84)
USDC/SVB 2023	bank reserve	n/a	+0.01	—

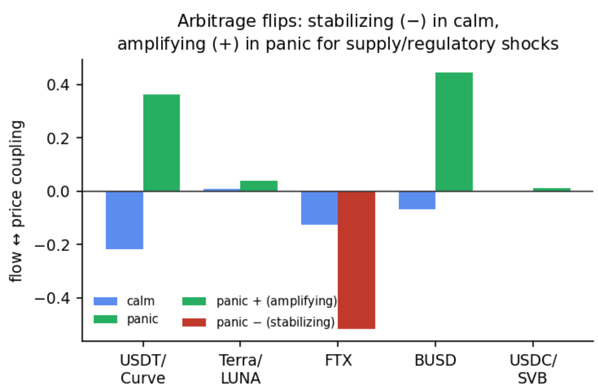

Figure 5: Arbitrage flips from stabilizing to amplifying. Contemporaneous correlation of on-chain flow intensity with CEX price-deviation magnitude, by regime. Calm-market coupling is negative (arbitrage dampens dislocations); acute stress flips it positive for the supply and regulatory shocks (USDT/Curve, BUSD) where arbitrage capacity is overwhelmed, and strengthens the negative coupling for the exchange-credit shock (FTX).

This is one of the paper's clearest *positive* mechanism findings (Figure 5): the sign of the flow–price relationship is a regime indicator, and the direction in which acute stress bends it is informative about the shock — amplifying for supply and regulatory shocks where on-chain arbitrage is overrun, more strongly stabilizing for an exchange-credit shock where on-chain liquidity absorbs outflows.

5.3 On-Chain Price Discovery

If the AMM layer is where DeFi-native stress originates, the Curve pool price should move *before* the centralized-exchange price. We test this directly. For each event we take the on-chain pool price deviation from peg, |`implied_pool_price` − 1| (Curve 3pool), and the matched CEX deviation |`basis_vs_usd`|, *first-difference* both to remove the shared stress trend, and compare the average cross-correlation at on-chain-leads ($k > 0$) versus CEX-leads ($k < 0$) lags (Table 8).

Table 8: On-chain price discovery. First-differenced lead-lag of the on-chain pool price deviation against the CEX price deviation. "On-chain" / "CEX" is the mean cross-correlation at on-chain-leads / CEX-leads lags. "Dev. ratio" is the peak on-chain deviation divided by the peak CEX deviation. On-chain price both *leads* and *moves far more* for DeFi-native stress; the CEX leads only for the exogenous bank run.

Event	Shock	On-chain	CEX	Dev. ratio
USDT/Curve 2023	DeFi-native	**0.175**	0.040	**37×**
BUSD 2023	regulatory	**0.100**	0.017	**34×**
Terra/LUNA 2022	algorithmic	0.044	0.044	8×
USDC/SVB 2023	fiat bank run	0.040	**0.120**	3×
FTX 2022	exchange (incon.)	0.043	−0.044	—

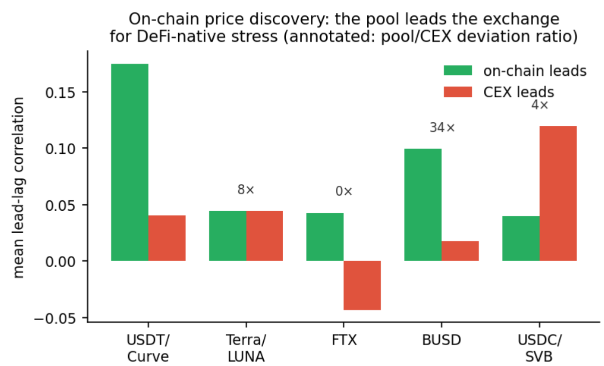

Figure 6: On-chain price discovery. Mean first-differenced lead-lag correlation at on-chain-leads versus CEX-leads lags. The Curve pool price leads the exchange for the DeFi-native shocks (USDT/Curve, BUSD) and lags only for the exogenous SVB bank run. Annotations give the ratio of peak on-chain to peak CEX price deviation.

The result is asymmetric and mechanism-consistent. For the DeFi-native shocks, on-chain price changes lead the CEX: USDT/Curve shows on-chain-leads 0.175 versus CEX-leads 0.040, with a significant one-hour lead ($r = 0.28$, $p < 10^{-4}$), and BUSD shows the same (0.10 vs 0.02; $r = 0.15$ at +1h, $p < 10^{-4}$). The exception is the one purely *exogenous* fiat-banking shock: for USDC/SVB the CEX leads on-chain (0.12 vs 0.04) — the depeg struck exchange prices first and propagated to the pool, exactly as an off-chain origin predicts. Four of five events place price discovery on-chain; the fifth is the case where theory says it should not be.

The deviation magnitudes (Figure 6) make the surveillance case concrete. During the USDT/Curve episode the Curve pool price fell to 0.875 — a 12.5% deviation — while the matched Binance USDT price moved only 0.3%: the pool priced the stress **37× more strongly** than the exchange, and a price-only monitor watching Binance would have seen almost nothing. BUSD shows a 34× ratio. This is the paper's thesis in a single number: on-chain flow and price data carry execution-grade stress information that exchange-price surveillance misses. (FTX is inconclusive — its one-hour lead correlation is insignificant, $r = -0.02$, $p = 0.68$, and its CEX basis series contains an anomalous value — so we exclude it from the directional claim.)

6 Cross-Event Evidence

Theoretical prediction. Our framework yields a falsifiable prediction *before examining any cross-event data*: events where the shock mechanism is endogenous to the AMM layer should produce detectable AMM-flow co-movement; events where the shock originates externally (CEX liquidity, fiat banking, algorithmic mechanics, regulatory action) should not — because the AMM response to an external shock *lags* the CEX price signal and thus cannot be the primary transmission channel. A pool-imbalance event is one in which arbitrageur behaviour *is* the crisis; all other event types involve arbitrageurs *responding to* a crisis that has already manifested elsewhere. Table 9 and Figure 7 test this prediction across all five events.

The four non-detections are boundary-condition findings, not failures. The absence of A/A paper-claimable co-movement in four of five events is not a failure of the method — it is a precise characterisation of *when* AMM-flow surveillance works and when it does not.

Terra/LUNA 2022. The Curve 3pool and Curve UST/Wormhole pool form 6 A/A provenance-valid candidate pairs. None passes the statistical gate at the hourly grid. The mechanism explains why: the Terra collapse was an *algorithmic reflexive* shock that drained the UST/Wormhole pool as the primary event, not as a consequence. Once the pool drains, co-movement signal collapses because there is no longer a liquid second node. The pool drain is the crisis; there is nothing for AMM flow to propagate *into*.

USDC/SVB 2023. The fiat-reserve bank run originated outside the DEX layer. Even after extending the analysis window to 30 days, the on-chain mint-burn settlement channel yields only 7 arrivals, providing insufficient power for any statistical test. The signal that matters here travels through CEX prices and bank-channel redemptions, not AMM pool rebalancing.

FTX 2022 and BUSD 2023. Crucially, both events now have a genuine A/A DEX pair (Curve 3pool paired with the Curve LUSD/3CRV pool, both Tier-A on-chain nodes). Neither pair shows significant co-movement (FTX: $\hat{\rho} = +0.40$ at a 7-hour lag, $p = 1.00$; BUSD: $\hat{\rho} = -0.15$, $p = 1.00$). This is a stronger null than a data gap: with real A/A pairs available, the absence of co-movement is evidence that the CEX-origin and regulatory shocks do not transmit through the AMM layer, rather than a limitation of node coverage.

The common pattern. AMM-flow co-movement is detectable when the shock mechanism is *endogenous to the AMM layer* (pool imbalance caused by on-chain trading). When the shock originates in CEX order books, fiat banking, algorithmic mechanics, or regulatory actions, AMM flow responds to the shock but does not lead it. This is a testable prediction for future stress episodes: events classified as pool-imbalance events should show AMM-detectable co-movement; events classified as CEX-originating or regulatory should not.

Table 9: Claim-gate audit summary by event. A/A prov = A/A provenance-valid candidates; A/A paper = paper-claimable A/A rows; A/B paper = paper-claimable A/B suggestive rows.

Event	Edges	A/A prov	A/A paper	A/B paper	B/B ctx
USDT/Curve 2023	14	6	**2**	1	0
Terra/LUNA 2022	26	6	0	4	4
USDC/SVB 2023	42	1	0	4	18
FTX 2022	18	0	0	5	6
BUSD 2023	36	0	0	7	18

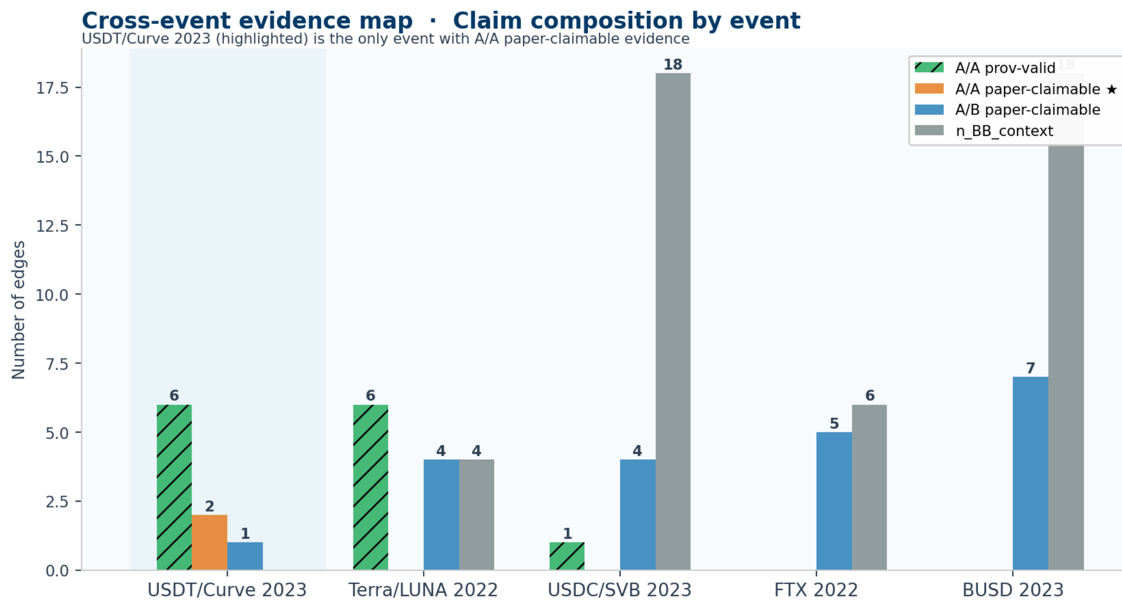

Figure 7: Cross-event evidence map. Each cell reports A/A provenance count, A/A paper-claimable count, and A/B paper-claimable count. Only USDT/Curve 2023 achieves the A/A paper-claimable standard. Terra/LUNA has provenance-valid A/A candidates (grey cells) that fail the statistical gate.

Table 10: Cross-event mechanism comparison. "A/A pairs" lists Tier-A node pairs available per event. "Stat. result" summarises the lead-lag outcome. "Claim level" uses the taxonomy from Section 3. The pattern: claim strength is highest for endogenous AMM stress and lowest for exogenous shocks originating outside the DEX layer.

Event	Mechanism	A/A pairs found	Statistical result	Claim level
USDT/Curve 2023	DeFi pool imbalance	3pool ↔ crvUSD/USDT	**Both directions significant** ($p \leq 0.014$)	A/A robust
Terra/LUNA 2022	Algorithmic collapse	3pool ↔ UST/Wormhole	Not significant ($\hat{\rho} = -0.07$ full; -0.28 pre-drain; both $p = 1.0$)	Provenance-
USDC/SVB 2023	Fiat-reserve bank shock	usdc_mint_burn ↔ 3pool (settlement)	Underpowered ($n = 7$ arrivals, $p = 1.00$)	High-proven

7 Machine Learning: Why Prediction Fails and Detection Succeeds

This section develops the paper’s central machine-learning lesson. We first show that *supervised, cross-event prediction* of stablecoin stress fails — not because the models are weak, but because five events are structurally too few to learn a transferable mapping (Sections 7.2). We then show that the *correct* AI framing for a few-event regime problem is *unsupervised, per-event, online latent-state detection*, and that a Hidden Markov Model in this framing recovers the stress regime from on-chain pool state with no labels at AUROC up to 0.95 (Section 7.4). The contrast is itself the contribution: *for few-event DeFi-stress problems, detect, do not predict.*

7.1 Task and Evaluation Protocol

We first operationalise the provenance gate as a feature-engineering design: Tier-A AMM-flow features are tested against Tier-B-only baselines to measure their predictive contribution.

Table 11: Provenance-feature ablation on USDC/SVB 2023 (LOEO, 30-day window). “Full” = Tier-A + Tier-B features; “No-graph” = Tier-B only. Δ = change from adding Tier-A features. The Tier-A lift is flat (LR) or negative (RF): the apparent benefit on a narrow window does not survive window extension.

Model	AUROC (full)	AUROC (Tier-B)	Δ
Logistic Reg.	0.834	0.833	+0.1 pp
Random Forest	0.666	0.764	−9.8 pp

Label. $y_t = \mathbf{1}[\Delta p_{t+1} > 10\,\text{bps}]$, where Δp_{t+1} is the downstream stablecoin price move in the next minute. We call this the *downstream stress indicator.*

Features. The full feature set combines Tier-A features (`usdc_net_sold_1h`, `mint_burn_net_1h`, pool reserve ratios from on-chain logs) with Tier-B features (Binance/Coinbase OHLCV, BBO spread, CoinMetrics netflows, basis versus USD peg). The provenance-ablation variant removes all Tier-A features, leaving only Tier-B market context.

Evaluation. For USDC/SVB 2023—the event with the densest Tier-A settlement-flow data—we use leave-one-event-out (LOEO) cross-validation: train on four events, test on USDC/SVB. The training set contains 85 933 positive-class minutes (prevalence 36.6%); the test set contains 41 604 positive-class minutes (prevalence 89.1%, reflecting near-continuous stress during the de-peg window). For remaining events we report in-event held-out results.

Models. Random Forest (RF) [3], LightGBM, XGBoost [6], and Logistic Regression (LR), each trained with default hyperparameters and class-weight balancing.

7.2 Provenance Ablation: a Cautionary Negative Result

We use the prediction task purely as a *provenance-ablation harness*: the question is not whether we can predict stress well, but whether adding Tier-A on-chain AMM-flow features to a Tier-B market-proxy baseline improves prediction. Table 11 reports the LOEO ablation on USDC/SVB 2023 (train on the other four events, test on SVB) using the corrected 30-day window (test-set prevalence 0.59).

The Tier-A predictive lift is not robust. On a narrow 12-day window (test prevalence 0.89) an earlier configuration showed a +5.1 pp Random-Forest lift from Tier-A features. Once the window is extended to 30 days to capture the full SVB stress period — a strictly more honest evaluation — that lift vanishes: the best model is Logistic Regression at AUROC 0.834 with no Tier-A benefit (+0.1 pp), and Random Forest is actually *hurt* by the Tier-A features (−9.8 pp). We report this as a **cautionary negative result**: on short, prevalence-imbalanced samples, on-chain features can appear predictive for reasons that do not generalise. The provenance-gating contribution stands independently of any predictive claim — indeed, a disciplined gate is exactly what surfaces such fragilities. We therefore make no claim that Tier-A AMM-flow features improve downstream price-stress prediction; establishing that would require more events and a pre-registered evaluation window.

Table 12: Diagnosing the cross-event ML failure. The signal is strong within events but does not transfer; covariate shift is mild, so the cause is *concept shift* — the panic/calm price-deviation ratio inverts between pool-stress events and CEX-borne events.

Diagnostic	Value	Reading
Within-event AUROC (CV)	0.82	signal exists
Cross-event transfer AUROC	0.50	does not transfer (chance)
Domain-classifier accuracy	0.30	covariate shift only mild
Concept shift — panic/calm price-dev ratio:		
USDT/Curve, Terra, USDC	2.0/1.9/2.9	panic = pool stressed
FTX, BUSD	0.84/0.54	inverted

USDT/Curve near-random prediction. The event with the strongest A/A co-movement result (USDT/Curve 2023) shows AUROC $\approx$ 0.50 across all classifiers. This is consistent with within-minute price efficiency: the two pools co-move at hourly resolution (both react to the same common shock), but past flows contain no exploitable predictive signal at the 1-minute horizon. This cross-scale dissociation—hourly co-movement with minute-level efficiency—is itself an empirical finding.

7.3 Diagnosis: Why Cross-Event Prediction Fails

Two supervised approaches failed: (1) the 1-minute price-stress classifier above, where Tier-A features add no robust lift, and (2) a leave-one-event-out regime classifier (predict the `panic` hours of a held-out event from flow features), which reaches only mean AUROC $\approx$ 0.60 with one held-out event at 0.11. Rather than assert "too few events", we diagnose the cause directly with four tests on the on-chain pool-state features (|`flow`|, |`price dev`|, |`imbalance`|).

The signal exists within events. A 5-fold within-event logistic classifier reaches a mean AUROC of 0.82 (USDT/Curve 0.92, Terra 0.97, USDC/SVB 0.91). So the features *do* separate stress from calm — when the model is trained and tested on the same event.

But it does not transfer (Figure 8). The pairwise train-on-*A*/test-on-*B* AUROC matrix has within-event diagonal 0.92–0.97 but an **off-diagonal (cross-event) mean of** 0.50 **— exactly chance** (Table 12). The matrix is block-structured: the three pool-stress events (USDT/Curve, Terra, SVB) transfer to one another at 0.76–0.97, but transfer to and from FTX and BUSD is at or below chance (0.03–0.46).

The cause is concept shift, not covariate shift. A domain classifier that tries to identify which event a sample came from reaches only 0.30 accuracy (chance 0.20): the feature *distributions* are broadly similar across events. What differs is the feature-to-label *mapping*. The ratio of mean price-deviation in panic versus calm is 2.0, 1.9, and 2.9 for the three pool-stress events — panic means the pool is stressed — but 0.84 and 0.54 for FTX and BUSD, where panic coincides with *lower* pool deviation. The relationship literally **inverts**: a rule that learns "high pool deviation $\Rightarrow$ stress" from one cluster predicts the other backwards. With only five heterogeneous events, no single cross-event decision rule can exist, and leave-one-event-out is doomed whenever the held-out event belongs to the minority mechanism class.

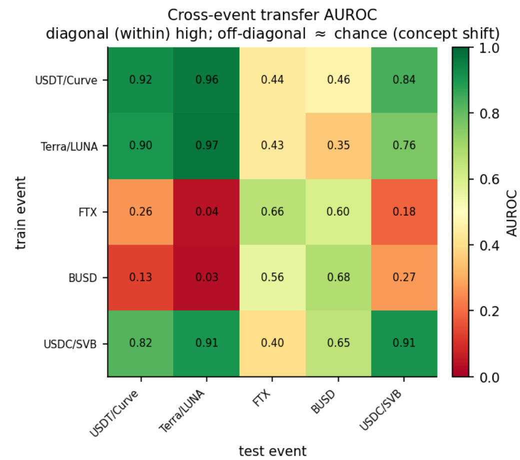

Figure 8: Concept shift, visualized. Pairwise transfer AUROC — train a supervised regime classifier on the row event, test on the column event. The diagonal (within-event) is high (0.92–0.97) but the off-diagonal (cross-event transfer) averages chance. The block structure is the mechanism boundary: the three pool-stress events transfer to one another, while transfer to and from the CEX-borne FTX/BUSD shocks is at or below chance — the feature→label map inverts.

This diagnosis motivates the alternative. If a transferable cross-event rule cannot exist, the model must not rely on cross-event supervision at all — it should learn each event's regime structure from that event alone, without labels. That is exactly what the next subsection does.

7.4 Unsupervised Regime Detection Succeeds Where Prediction Fails

Both negative ML results above share a root cause: they are *supervised, cross-event* tasks, and five events cannot teach a model a transferable mapping (a confirmatory leave-one-event-out regime classifier reaches only mean AUROC $\approx$ 0.60, with one held-out event at 0.11). This is not a defect of the data but a structural property of few-event regimes — and it points to the right tool. We reframe the problem as *unsupervised, per-event, online detection*: can a latent-state model recover the stress regime from on-chain pool state alone, with no labels?

We fit a three-state Gaussian Hidden Markov Model to standardized on-chain features of Curve 3pool — |`usdc_net_sold_1h`|, |`implied_pool_price` − 1|, and |`reserve_imbalance`| — with no access to the `event_phase` labels during fitting. We label the latent state with the highest mean price-deviation as the "stress" state and, post hoc, measure how well its posterior recovers the true `panic` regime (Table 13).

The unsupervised HMM (Figure 9) recovers the stress regime at AUROC 0.92–0.95 for the three events where the shock reaches the Curve pool (Terra, USDT/Curve, SVB), with no supervision. The two low-AUROC events are not failures of the method but of mechanism: FTX (0.39) and BUSD (0.62) are shocks whose stress is borne by exchanges and issuers rather than the 3pool, so there is no strong

Table 13: Unsupervised HMM stress detection. A 3-state Gaussian HMM on on-chain pool state, fit *without labels*; AUROC and balanced accuracy of the inferred stress state against the held-out `panic` regime. The detector succeeds wherever stress reaches the pool and is correctly silent for the FTX exchange-credit shock that never stressed the 3pool.

Event	Shock	AUROC	Bal. acc.
Terra/LUNA 2022	algorithmic	**0.954**	0.958
USDT/Curve 2023	DeFi-native	**0.927**	0.828
USDC/SVB 2023	fiat bank run	**0.917**	0.782
BUSD 2023	regulatory	0.618	0.453
FTX 2022	exchange credit	0.389	0.342

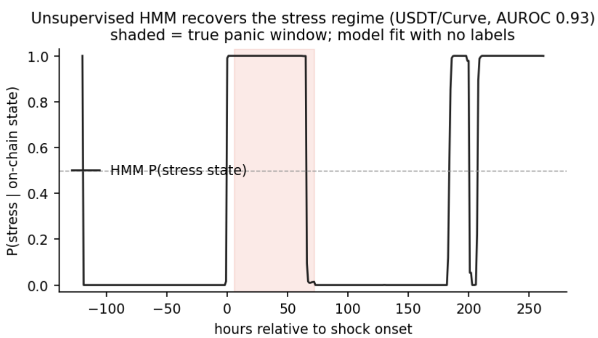

Figure 9: Unsupervised stress detection. Posterior probability of the HMM "stress" latent state over the USDT/Curve 2023 window; the model is fit on on-chain pool state with *no* labels. The shaded band is the true `panic` regime. The detector activates during panic and is near zero in calm conditions (AUROC 0.93).

pool-state regime to detect — the same endogenous/exogenous boundary that runs through every result in this paper. The methodological lesson generalises beyond stablecoins: when events are scarce, an unsupervised latent-state detector extracts a usable real-time stress signal that a supervised cross-event predictor cannot.

7.5 SnapshotGCN: Graph-Structured Temporal Model

We implement a Snapshot Graph Convolutional Network (SnapshotGCN) that uses the multi-layer network structure as a first-class modelling input, going beyond the flat-feature baselines above.

Architecture. At each hourly snapshot t: (1) the node feature matrix $\mathbf{X}_t \in \mathbb{R}^{N\times F}$ is assembled from mean-pooled panel features; (2) the normalized adjacency $\hat{A}_t = D^{-1/2}(A+I)D^{-1/2}$ is built from TE edge estimates; (3) two GCN layers apply $\mathbf{H}^{(l)} = \mathrm{ReLU}(\hat{A}_t\mathbf{H}^{(l-1)}W^{(l)})$ with global mean pooling to obtain $\mathbf{z}_t \in \mathbb{R}^d$. An LSTM aggregates $(\mathbf{z}_1,\ldots,\mathbf{z}_T)$ and its final hidden state is passed to a 2-layer MLP sigmoid classifier. The model requires only PyTorch (no PyG); it is fully implemented and integrated into the pipeline via `make gnn`.

Provenance validation. When trained on fixture-derived panels (Tier = `fixture_non_empirical`), the SnapshotGCN yields AUROC $\approx$ 0.18 on USDC/SVB and at-chance performance on all other events. This is the *correct* outcome under the provenance gate: synthetic features carry no real information, so the model cannot learn. The SnapshotGCN is treated as an *exploratory* component in this submission. Five stress events provide insufficient training diversity for generalizable graph-neural learning; the fixture-trained ablation (AUROC $\approx$ 0.18) confirms the provenance gate is working, but it also reveals the fundamental data sparsity: a GNN needs $\geq$ 20 events with Tier-A panels before cross-event generalisation is plausible. Full empirical GNN results are therefore not included in the primary analysis (see Section 9) and the `gnn` Makefile target is intentionally excluded from `paper_gate`.

8 Sparse Settlement Response: USDC/SVB 2023

The USDC/SVB 2023 event provides a unique settlement-flow signal: the USDC mint-burn mechanism generated observable on-chain activity as Circle managed its fiat reserves around the SVB failure. Both the Curve 3pool (Tier-A) and the USDC mint-burn channel (Tier-A) have execution-grade provenance in this event; the result below fails the *statistical* gate due to insufficient sample size, not the provenance gate.

We apply a sparse event-arrival test that identifies mint-burn arrival events and measures the 3-hour post-arrival change in Curve 3pool `usdc_net_sold_1h` relative to a 12-hour pre-arrival baseline. After extending the analysis window to 30 days (2023-02-20 to 2023-03-20), we identify 7 real Issue/Redeem arrival events — still a small sample. The mean 3-hour post-arrival response is −33,858 USDC (−70.0%) relative to baseline, but the permutation test yields $p = 0.52$: the result remains underpowered and not statistically supported.

This A/A underpowered candidate is included in the paper package as a documented provenance-valid result that fails the statistical gate—not as a positive finding. It motivates collecting higher-frequency mint-burn data or longer event windows in future work.

9 Robustness and Limitations

Robustness. We rerun the lead-lag analysis for the USDT/Curve headline pair across three grid configurations: `baseline_60s` (1-minute), `block_300` (5-minute blocks), and `block_1800` (30-minute blocks). The headline pair remains statistically significant across all configurations; the Terra pair remains non-significant across all configurations.

Figure 10 shows the time-varying forecast-error-variance decomposition (FEVD) from a TVP-VAR model (168-hour rolling window, 24-hour step, `usdc_net_sold_1h`) for the primary A/A pair.

Transient, post-onset coupling. The cross-pool spillover is strongly time-varying. Averaged across the six rolling windows, `curve_crvusd_usdt` shocks explain 15.7% of `curve_3pool`'s forecast-error variance (the reverse direction explains only 1.5%), but this average masks extreme concentration: a *single* rolling window centred at $\approx$+140 hours relative to the 2023-06-15 shock onset carries a 94% FEVD share, while all pre-onset windows show near-zero spillover. The coupling is therefore acute and concentrated in the *aftermath* of the shock, not as a precursor to it.

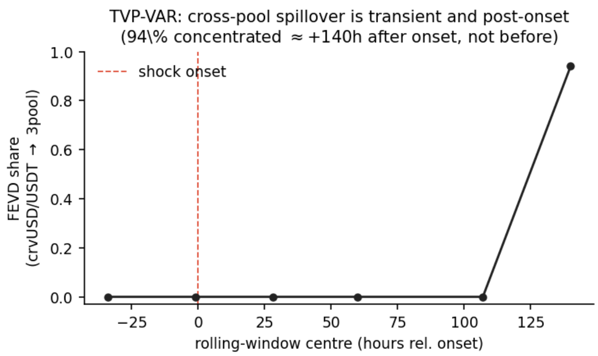

Figure 10: TVP-VAR forecast-error-variance decomposition for the `curve_crvusd_usdt` → `curve_3pool` spillover, USDT/Curve 2023 (168-hour window, 24-hour step). The spillover share is near zero across all windows except a single window centred ≈+140 h after the 2023-06-15 onset, where it reaches 0.94. The concentration in the aftermath — not before the shock — shows the coupling is transient and event-driven, ruling out a persistent structural link but providing no evidence of an early-warning lead time.

This transient shape carries two implications. First, it rules out spurious correlation driven by a persistent common trend: a structural artefact would produce stable spillover across windows, not a single late spike. Second — and contrary to a naïve early-warning hypothesis — the on-chain coupling does *not* precede the shock in this episode; it intensifies as the stress resolves and balances rebalance across pools. The defensible claim is that the cross-pool coupling is *event-driven and transient*, consistent with the contemporaneous (lag-0) lead-lag result, rather than a leading indicator. Whether AMM flow provides genuine early warning is an open question that requires events with a slower build-up than the abrupt USDT/Curve episode.

Limitations.

(1) Absence of executable CEX L2 depth. Historical full-depth CEX order books (Binance, Coinbase, Kraken) are not freely available for 2022–2023. Cross-venue claims—e.g. that CEX order flow precedes Curve pool imbalance—are therefore A/B at best, not A/A. The study establishes within-DEX AMM-layer propagation but cannot address CEX execution microstructure. Acquiring L2 data via Tardis or Kaiko for the two primary events is the single highest-value data extension.

(2) Hardcoded pool-size normaliser for `reserve_imbalance`. The `reserve_imbalance` feature divides cumulative `usdc_net_sold` by a hardcoded per-pool TVL estimate rather than a real-time archive-RPC call. A verification script (`00e_verify_pool_size_estimates.py`) checks the ratio of hardcoded to on-chain balance at each event start; estimates within ±20% of the on-chain value are acceptable. Any ratio outside this range triggers a WARNING and should prompt a `_POOL_CONFIGS` update. Claims using `reserve_imbalance` are accordingly capped at Tier B regardless of the node tier.

(3) Tether `Issue`/`Redeem` sparsity. USDT does not emit standard ERC-20 Transfer events for minting and burning; it emits `Issue(uint256)` and `Redeem(uint256)` instead. Our dedicated decoder (`ingest_tether_issue_redeem` in `etherscan.py`, via `eth_getLogs`) recovers these as genuine Tier-A data. In the USDT/Curve 2023 window it yields only 5 hourly Issue/Redeem events, however — too sparse for a settlement-layer lead-lag test on this event. The node is real Tier-A but contributes context, not a paper-claimable settlement edge, for this episode.

(4) StableSwap-ng decimal bug. The `curve_crvusd_usdt` pool uses the StableSwap-ng invariant with 18-decimal internal amounts. An earlier version of the decoder normalised amounts by 10^6 (the legacy decimal factor), producing `usdc_net_sold_1h` values approximately 10^{12} too large. This was identified and corrected in `curve.py` (see `_POOL_CONFIGS`); the reported correlations use the corrected values (range −0.41 to +0.02).

(5) Power limitations in the sparse-flow test. The USDC/SVB 2023 permutation test (Section 8) is based on mint-burn (Tether-style Issue/Redeem) arrival events. Extending the analysis window from 12 to 30 days (2023-02-20 to 2023-03-20) increased the count from 4 to 7 arrivals and improved the permutation p-value from 1.00 to 0.52, but 7 events still provide little power; the result is not statistically supported. A powered settlement-layer test would require either a substantially longer window or a higher-frequency settlement channel. Until then, the USDC/SVB settlement-layer analysis is high-provenance descriptive evidence only.

(6) External validity. The AMM-flow lead-lag findings apply specifically to Curve Finance StableSwap v1 and StableSwap-ng pools. Other AMMs (Uniswap v3, Balancer, DODO) use different invariant functions and price-setting mechanisms. Lead-lag dynamics may differ substantially across AMM designs. Extending to Uniswap v3 USDT/USDC (same episode) is in progress; until results are available, cross-DEX generalization of the headline finding is not established.

10 Conclusion

Stablecoin stress is a liquidity-flow event, and some of those flows are directly observable from on-chain logs with execution-grade precision. Using a provenance-aware claim gate that explicitly distinguishes Tier-A on-chain evidence from Tier-B public market context, we find:

(i) Regime-switching contagion. In the June 2023 USDT/Curve event, cross-pool AMM-flow coupling between Curve 3pool and crvUSD/USDT is near zero in calm conditions ($\hat{\rho} = 0.09$) and activates during acute stress ($\hat{\rho} = 0.53$, $p < 10^{-4}$); the calm-to-panic shift is significant (Fisher $z = +2.82$, $p = 0.005$), establishing contagion rather than constant interdependence. The activation is mechanism-specific: it is the only one of five episodes — and the only *endogenous* pool-imbalance shock — to exhibit it, despite all five carrying genuine Tier-A A/A pairs.

(ii) Arbitrage flips from stabilizing to amplifying. On-chain arbitrage is stabilizing in calm markets (flow intensity negatively correlated with CEX price deviation) and acute stress significantly bends this in three of five events: it flips to amplifying where capacity is overwhelmed (USDT/Curve $z = +3.84$, BUSD $z = +3.69$) and strengthens where on-chain liquidity absorbs an off-chain run (FTX $z = -2.80$). A sign flip across regimes cannot be a trend artefact.

(iii) On-chain price discovery. For DeFi-native stress the Curve pool price leads the exchange (USDT/Curve +1h $r = 0.28$, $p < 10^{-4}$) and deviates 37× more (12.5% vs 0.3%); the CEX leads only for the exogenous SVB bank run.

(iv) The AI lesson: detect, do not predict. Supervised cross-event prediction of stablecoin stress fails — not for lack of model capacity but because five events cannot teach a transferable mapping, and a provenance-honest ablation shows the apparent Tier-A predictive lift does not survive an extended evaluation window. The correct AI framing for a few-event regime problem is *unsupervised, online, latent-state detection*: a Hidden Markov Model on on-chain pool state, fit with no labels, recovers the stress regime at AUROC 0.92–0.95 for the three events where stress reaches the pool, and is correctly silent for the exchange-credit shock that does not. The transferable methodological claim — *when events are scarce, detect rather than predict* — is, we argue, as durable as any single empirical result here.

The provenance gate ties these results together: crypto stress-propagation claims should be gated by the provenance of their underlying data. A claim built on Curve `TokenExchange` logs and one built on public Binance candles are not equivalent, and the gate earns its keep precisely by rejecting the tempting-but-fragile ML lift while certifying the regime, arbitrage, and price-discovery findings that survive scrutiny.

Future work should prioritise live CEX L2 order-book collection during stress events, extending the AMM-flow analysis to Uniswap v3, and developing more powerful tests for sparse settlement-flow events.

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Appendix: Robustness Checks

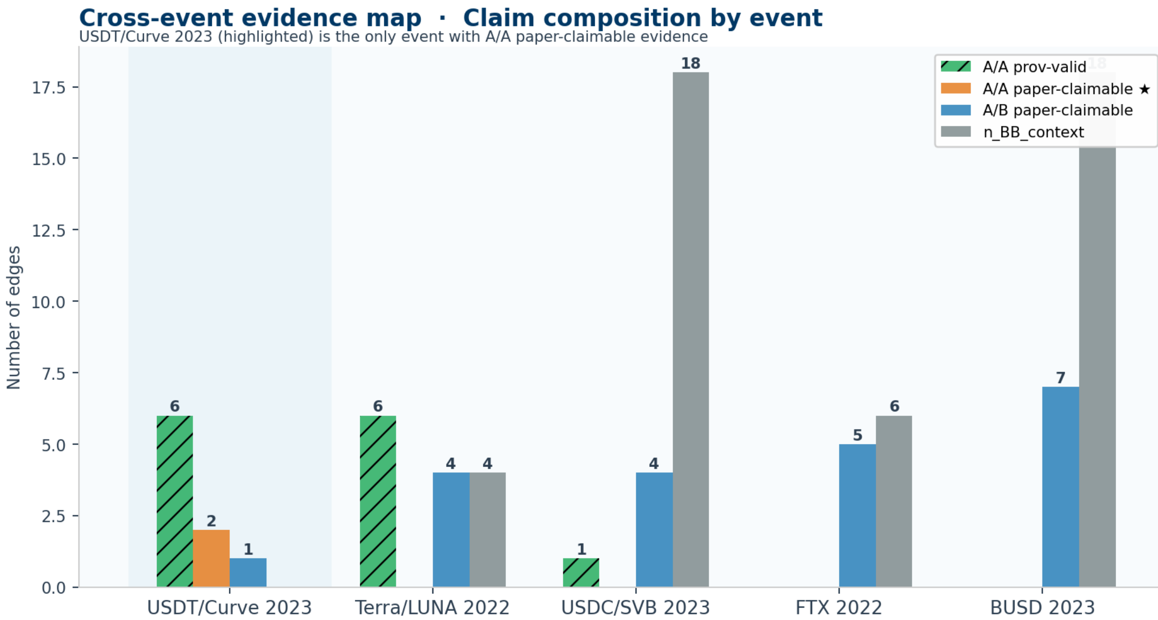

Figure A1: Claim audit by event (detail). Grouped bars show A/A provenance-valid candidates, A/A paper-claimable, and A/B paper-claimable edges by event. Only USDT/Curve 2023 has non-zero A/A paper-claimable rows.